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Quantitative Aptitude

Formula – Guide

Logarithm

Logarithm

Introduction to Logarithms

- If $a^x = m$, then x is the logarithm of m to base a, written as $\log_a m = x$.

Key Properties of Logarithms

- Logarithm to any base of 1 is 0: $\log_a 1 = 0$ for $a > 0, a \neq 1$

  $a^0 = 1$, hence $\log_a 1 = 0$.
- The logarithm of a number to the same base is 1: $\log_a a = 1$ for $a > 0, a \neq 1$

  $a^1 = a$, hence $\log_a a = 1$.

Common and Natural Logarithms

- **Common logarithms:** Base 10 logarithms, denoted as $\log_{10} m$ or simply $\log m$ (assumed base 10 if no base is mentioned).
- **Natural logarithms:** Logarithms to base e (Euler's number), denoted as $\ln m$.

Laws of Logarithms

- **Product Rule:** $\log_a(mn) = \log_a m + \log_a n$
- **Quotient Rule:** $\log_a\left(\frac{m}{n}\right) = \log_a m - \log_a n$
- **Power Rule:** $\log_a(m^n) = n \cdot \log_a m$
- **Base Change Rule:**

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Logarithmic Inequalities

- If $a > 1$, then $\log_a x_1 > \log_a x_2$ implies $x_1 > x_2$.
- If $a < 1$, then $\log_a x_1 > \log_a x_2$ implies $x_1 < x_2$.

Applied Results in Logarithms

- The characteristic of common logarithms of any positive number less than 1 is negative.
- The characteristic of common logarithms of any number greater than 1 is positive.
- If the logarithm to any base gives the characteristic n, then the number of integers is given by $a^{n+1} - a^n$.

Logarithm Rules Table

Rule Name	Formula
Log of 1	$\log_b 1 = 0$
Log of the same number as the base	$\log_b b = 1$
Product Rule	$\log_b (xy) = \log_b x + \log_b y$
Quotient Rule	$\log_b (x/y) = \log_b x - \log_b y$
Power Rule	$\log_b (ax) = x * \log_b a$
Change of Base Rule	$\log_a b = \log_c a / \log_c b$ OR $\log_a b = \log_c b - \log_c a$
Equality Rule	$\log_b a = \log_b c \rightarrow a = c$
Other Rule	$\log_b (mn) = m/n * \log_b a$ OR $-\log_b a = \log_b (1/a)$ $\log_1 b = \log_b a$

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